Sección 10.2

3) $x = te^t$
$y = t + \operatorname{sen} t$

$\frac{dx}{dt} = 1e^t + te^t$
$e^t + e^t t$

$\frac{dy}{dt} = 1 + \cos t$

$\frac{dy}{dx} = \frac{1+\cos t}{e^t + e^t t}$

5) $x = t^2 + 2t$
$y = 2^t - 2t$

$\frac{dy}{dt} = 2^t \ln(2) - 2$ $\quad \frac{dx}{dt} = 2t + 2$

$m = \frac{2^t \ln(2) - 2}{2t + 2} = \frac{2^3 \ln(2) - 2}{2(3)+2} = \underline{0.443147}$

$15 = t^2 + 2t$
$2 = 2^t - 2t$

$t^2 + 2t - 15 = 0$
$(t+5)(t-3) = 0$
$t = -5$
$t = 3$

$2^t - 2t - 2 = 0$

9) $x = \operatorname{sen} 2t + \cos t$ $\quad t = \pi$
$y = \cos 2t - \operatorname{sen} t$

$\frac{dy}{dt} = 2\operatorname{sen} 2t - \cos t$

Si $t = \pi$

$x = \operatorname{sen} 2\pi + \cos \pi = -1$
$y = \cos 2\pi - \operatorname{sen} \pi = 1$

$\frac{dx}{dt} = -2\operatorname{sen} 2t - \operatorname{sen} t$

$m = \frac{2\operatorname{sen} 2t - \cos t}{2\operatorname{sen} 2t - \operatorname{sen} t} = \frac{2\operatorname{sen} 2\pi - \cos \pi}{2\operatorname{sen} 2\pi - \operatorname{sen} \pi} = \frac{1}{2}$

$y - 1 = \frac{1}{2}(x+1)$
$y - 1 = \frac{1}{2}x + \frac{1}{2}$

$\underline{y = \frac{1}{2}x + \frac{3}{2}}$

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11) $x = \operatorname{sen} t$
$y = \cos^2 t$

$\frac{1}{2} = \operatorname{sen} t$

$\operatorname{sen}^{-1}\left(\frac{1}{2}\right) = t$

$t = \frac{\pi}{6}$

$\frac{3}{4} = \cos^2 t$

$\frac{dy}{dt} = 2\cos t \cdot -\operatorname{sen} t = 2co$
$-\operatorname{sen} 2t$

$\frac{dx}{dt} = \cos t$

$m = \frac{-\operatorname{sen} 2t}{\cos t} = \frac{-\operatorname{sen} 2\frac{\pi}{6}}{\cos \frac{\pi}{6}} = -1$

$y - \frac{3}{4} = -1\left(x - \frac{1}{2}\right)$

$y - \frac{3}{4} = -x + \frac{1}{2}$

$y = -x + \frac{5}{4}$

13) $x = t^2 - t$
$y = t^2 + t + 1$

$(0,3)$

$0 = t^2 - t$
$0 = t(t-1)$
$t = 0$
$t = 1$

$3 = t^2 + t + 1$
$0 = t^2 + t - 2$
$t = -2$
$t = 1$

$\frac{dx}{dt} = 2t - 1$

$\frac{dy}{dt} = 2t + 1$

$m = \frac{2t+1}{2t-1} = \frac{2(1)+1}{2(1)-1} = \frac{3}{1} = 3$

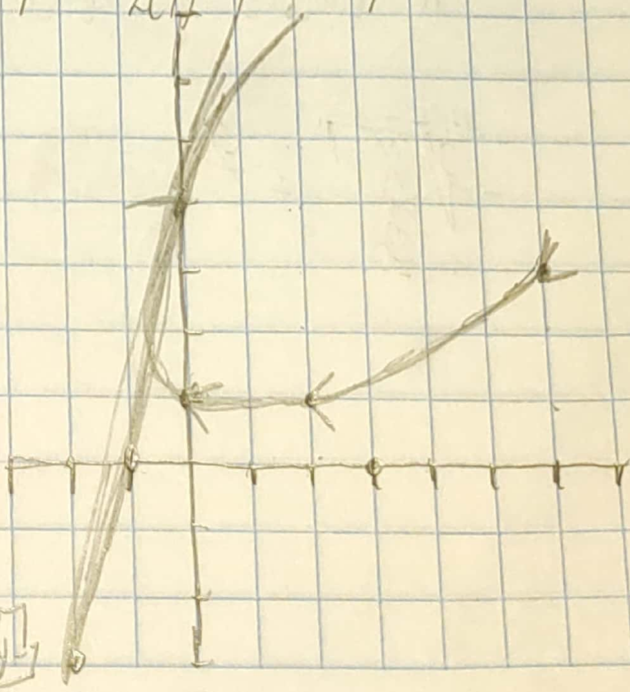
$y - 3 = 3(x - 0)$
$y - 3 = 3x$
$y = 3x + 3$

t	x	y
-2	6	3
-1	2	1
0	0	1
1	0	3
2	2	7

$y - 1 = t^2 + t$

$\frac{\sqrt{y-1}}{2} = t$

$x = \left(\frac{\sqrt{y-1}}{2}\right)^2 - \frac{\sqrt{y-1}}{2}$



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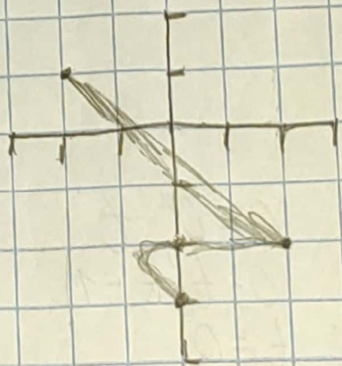
19) $x = t - \ln t$
$y = t + \ln t$

$\frac{dx}{dt} = 1 - \frac{1}{t}$ $\qquad$ $\frac{dy}{dt} = 1 + \frac{1}{t}$

$m = \frac{1 + \frac{1}{t}}{1 - \frac{1}{t}} = \frac{\frac{t+1}{t}}{\frac{t-1}{t}} = \frac{t+1}{t-1}$ $\qquad$ $\frac{1(t+1) - (t-1)1}{(t-1)^2} = -\frac{2}{(t-1)^2}$

$\frac{d^2y}{dx^2} = \frac{-\frac{2}{(t-1)^2}}{1 - \frac{1}{t}} = \frac{-2}{(t-1)^2\left(1 - \frac{1}{t}\right)} = \frac{-2}{\frac{(x-1)^3}{x}} = \frac{-2t}{(t-1)^3}$

21) $x = t^3 - 3t$
$y = t^2 - 3$



$H = (0, -3)$

$V = (\pm 2, -2)$

t	x	y
-3	-18	
-2	-2	1
-1	2	-2
0	0	-3
1	-2	-2
2	2	1
3		

$\sqrt{y+3} = t$

$x = \left(\sqrt{y+3}\right)^3 - 3\left(\sqrt{y+3}\right)$

$-\sqrt[3]{\frac{x}{3}} = t \quad -t$

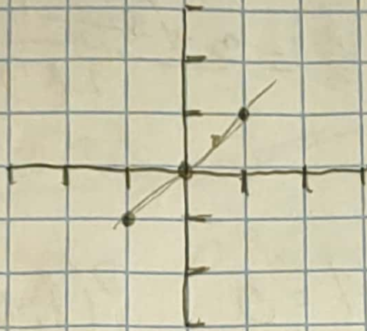
23) $x = \cos\theta$

$y = \cos 3\theta$

$\cos^{-1} x = \theta$

$y = \cos 3 \cos^{-1} x$

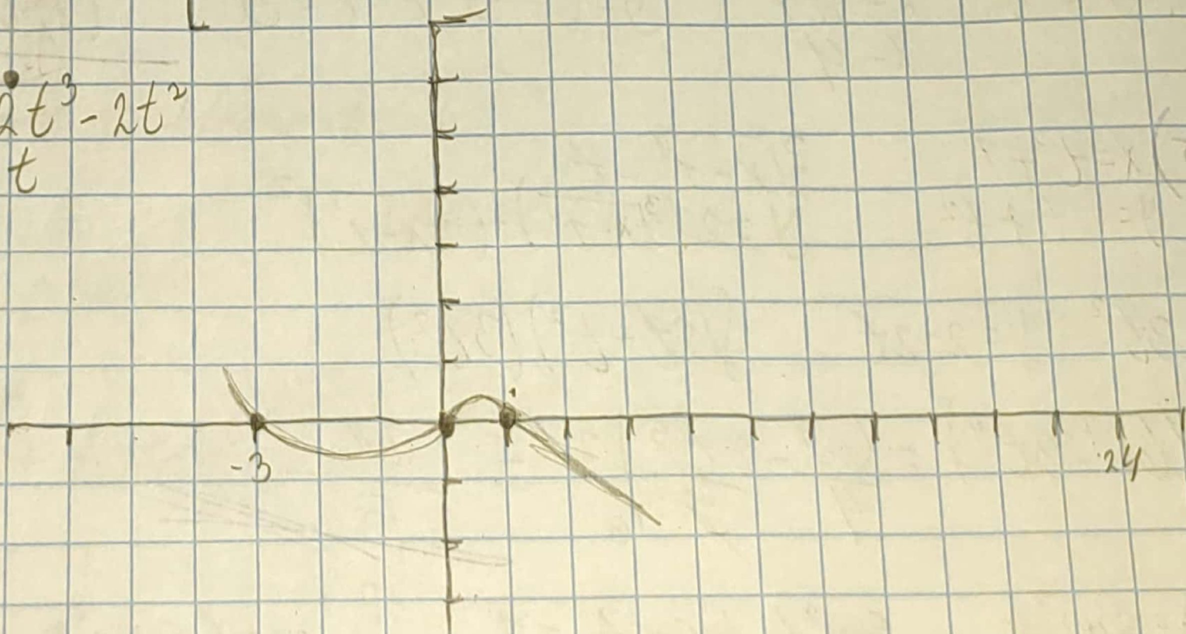
t	x	y
$-\pi$	-1	-1
$-\frac{\pi}{2}$	0	0
0	1	1
$\frac{\pi}{2}$	0	0
π	-1	-1



27) $x = t^4 - 2t^3 - 2t^2$

$y = t^3 - t$

t	x	y
-1	1	0
-2	24	-6
0	0	0
1	-3	0
2	-8	6



31) $x = r\theta - d \operatorname{sen}\theta$

$y = r - d\cos\theta$

$$\frac{r - d \operatorname{sen}\theta}{r - d\cos\theta}$$

Duda

33) $x = 3t^2 + 1$
$y = t^3 - 1$

$\sqrt{\frac{x-1}{3}} = t$

$y = \left(\sqrt{\frac{x-1}{3}}\right)^3 - 1$ $\quad m = \frac{3\left(\frac{x-1}{3}\right)^{1/2}}{2} \cdot \frac{1}{3} = \frac{3\sqrt{\frac{x-1}{3}}}{2 \cdot 3} = \frac{1}{2}$

$m = 1/2$

$x - 1 = 3$ $\quad x = 3 + 1$ $\quad y = \left(\sqrt{\frac{4-1}{3}}\right)^3 - 1 = 0$ $\quad P(4, 0)$
$x = 4$

35) $x = t^3 + 1$ $\quad \sqrt[3]{x-1} = t$
$y = 2t + t^2$ $\quad y = 2\left(\sqrt[3]{x-1}\right) - \left(\sqrt[3]{x-1}\right)^2$

$x = 3t^2$ $\quad y = 2 - 2t$ $\quad \int (2t - t^2)(3t^2)$

$\int_0^2 (6t^2 - 3t^4) = \frac{6t^4}{4} - \frac{3t^5}{3}\Big|_0^2 = \frac{24}{5}$

37) $x = sen^2 t$ $\quad sen^2 t + cos^2 t = 1$ $\quad \int_0^1 cost\ sen\theta =$
$x = cost$ $\quad x + y^2 = 1$

Duda

39) $x = a\cos\theta$ $\quad 0 \leq \theta \leq 2\pi$ $\quad \frac{x}{a} = \cos\theta \quad \frac{y}{b} = \operatorname{sen}\theta$
$y = b\operatorname{sen}\theta$

$\cos^2\theta + \operatorname{sen}^2\theta = 1 \quad \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$

$A = \pi r^2 = \underline{\pi ab}$

49) $x = t\operatorname{sen}t$ $\quad 0 \leq t \leq 1$
$y = t\cos t$

$\int_0^1 \sqrt{(\cos t - t\operatorname{sen}t)^2 + (\operatorname{sen}t + t\cos t)^2} =$

$y' = t\cos t = 1 \quad \cos t + t - \operatorname{sen}t$
$= \cos t - t\operatorname{sen}t$

$x' = t\operatorname{sen}t \quad 1\operatorname{sen}t + t\cos t$
$y' = \operatorname{sen}t + t\cos t$

55) $x = \operatorname{sen}^2 t$ $\quad 0 \leq t \leq 3\pi$
$y = \cos^2 t$

$L = \int_0^{3\pi} \sqrt{(\operatorname{sen}2t)^2 + (-\operatorname{sen}2t)^2} = \int_0^{3\pi} \sqrt{2\operatorname{sen}^2 2t} = \sqrt{2}\int_0^{3\pi} \sqrt{\operatorname{sen}^2 2t}\, dt$

$\sqrt{2}\int_0^{3\pi} \operatorname{sen}2t = -1.22 \times 10^{-14}$

57) $x = 2t - 5$
$y = 2t^2 - 3t + 6$

$\frac{x+3}{2} = t \quad y = 2\left(\frac{x+3}{2}\right)^2 - 3\left(\frac{x+2}{2}\right) + 6$